

Program : Diploma in Chemical Engineering	
Course Code : 6079	Course Title: Fuels and Combustion Lab
Semester : 6	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To provide hands on experience in the analysis of liquid, gaseous and solid fuels

Course Pre-requisites:

Topic	Course code	Course name	Semester
Analysis of petroleum products		Petroleum Refining & Petrochemical Technology	5

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Analyse liquid fuels.	24	Applying
CO2	Analyse gaseous fuels.	3	Applying
CO3	Analyse solid fuels	9	Applying
CO4	Demonstrate ASTM method of distillation of petroleum products.	9	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Analyse liquid fuels		
M1.01	Determination of flash point of solid fuel.	3	Applying
M1.02	Determination of fire point of solid fuel.	3	Applying
M1.03	Carbon residue test for solid fuels.	3	Applying
M1.04	Estimation of viscosity of lubricants.	3	Applying
M1.05	Estimation of cloud point.	3	Applying
M1.06	Estimation of pour point.	3	Applying
M1.07	Determination of Calorific value of liquid Fuel using Bomb Calorimeter.	3	Applying
M1.08	Open ended experiment	3	Applying
CO2	Analyse gaseous fuels		
M2.01	Determination of Calorific value of a fuel using Junkers Gas Calorimeter	3	Applying
	Series Test – I		
CO3	Analyse solid fuels		
M3.01	Proximate analysis of coal.	6	Applying
M3.02	Open ended experiment	3	Applying

CO4	Demonstrate ASTM method of distillation of petroleum products		
M4.01	ASTM distillation of kerosene.	3	Applying
M4.02	ASTM distillation of gasoline.	3	Applying
M4.03	Open Ended Experiment	3	Applying
	Series Test – II		

Text / Reference

T/R	Book Title/Author
T1	Bulk Chemicals from Petroleum; B.K. Bhaskarao,
R1	Modern Petroleum Refining Processes; B. K. Bhaskara,
R2	Petroleum Refinery Engineering; W.L. Nelson,
R3	Elements of Petroleum Refinery Engineering; O.P. Gupta,
R4	Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/105/103105110/
2	https://nptel.ac.in/courses/103/102/103102022/